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A social network dataset is a categorical dataset to determine whether a user purchased a particular product based on gender, age and estimated salary.

Objectives:

1. Understand the Dataset & cleanup (if required).
2. Use a SVM to classify whether a user purchased a car or not? (Use Linear Kernel)
3. Create a confusion matrix and evaluate the model using accuracy

```
import pandas as pd
# Load Dataset
df = pd.read_csv('social_network.csv')
print(df.head())
```

```
   Unnamed: 0  Gender  Age  EstimatedSalary  Purchased
0           0    Male   49           131217          1
1           1  Female   56            92991          1
2           2    Male   49            24014          0
3           3    Male   21            31093          0
4           4    Male   47            38070          0
```

```
df.drop(columns=['Unnamed: 0'], inplace=True)
print(df.head())
```

```
   Gender  Age  EstimatedSalary  Purchased
0    Male   49           131217          1
1  Female   56            92991          1
2    Male   49            24014          0
3    Male   21            31093          0
4    Male   47            38070          0
```

```
# Data Cleaning
df.isnull().sum()
```

```
Gender             0
Age                0
EstimatedSalary    0
Purchased          0
dtype: int64
```

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['Gender']=le.fit_transform(df['Gender'])
```

```
df.head()
```

```
   Gender  Age  EstimatedSalary  Purchased
0       1   49           131217          1
1       0   56            92991          1
2       1   49            24014          0
3       1   21            31093          0
4       1   47            38070          0
```

```
X = df[['Gender', 'Age', 'EstimatedSalary']]
y = df['Purchased']

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

print('X_train :\n', X_train)
```

```
X_train :
      Gender  Age  EstimatedSalary
79         0   26            74021
197        0   20            38071
38         1   54            90031
24         0   45            49592
122        0   20            40056
..       ...  ...              ...
106        0   59            28308
14         0   39           125864
92         0   24            37019
179        1   23           120793
102        0   42           132566

[160 rows x 3 columns]
```

```
print('X_train shape:\n', X_train.shape)
```

```
X_train shape:
 (160, 3)
```

```
print('X_test :\n', X_test.head())
print('X_test shape:\n', X_test.shape)
```

```
X_test :
      Gender  Age  EstimatedSalary
95         0   57           111794
15         1   45            98069
30         1   30           109912
158        1   39           141305
128        0   25            33500
X_test shape:
 (40, 3)
```

```
from sklearn.preprocessing import StandardScaler

# Feature scaling
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

X_train.shape, X_test.shape
```

```
((160, 3), (40, 3))
```

```
X_train
```

```
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```

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```

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```

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       [-1.03823026,  0.15901713,  1.27837906]])
```

```
from sklearn.svm import SVC

svm_model = SVC(kernel='linear', random_state=42) # to use RBF replace
with 'rbf'
svm_model.fit(X_train, y_train)
```

```
SVC(kernel='linear', random_state=42)
```

```
y_pred = svm_model.predict(X_test)

from sklearn.metrics import confusion_matrix, accuracy_score,
precision_score, recall_score, f1_score

# Confusion Matrix and Accuracy
cm = confusion_matrix(y_test, y_pred)
acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred)
rec = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)


print("\nConfusion Matrix:")
print(cm)
print(f"\n Model Accuracy: {acc*100:.2f}%")
```

```python
print(f"\n Model Precision: {prec*100:.2f}%")
print(f"\n Model Recall: {rec*100:.2f}%")
```

```
Confusion Matrix:
[[ 6  5]
 [ 6 23]]

 Model Accuracy: 72.50%

 Model Precision: 82.14%

 Model Recall: 79.31%
```

```python
import numpy as np
# Compute Confusion Matrix manually
TP = TN = FP = FN = 0
for actual, pred in zip(y_test, y_pred):
    if actual == 1 and pred == 1:
        TP += 1
    elif actual == 0 and pred == 0:
        TN += 1
    elif actual == 0 and pred == 1:
        FP += 1
    elif actual == 1 and pred == 0:
        FN += 1

cm = np.array([[TN, FP],
               [FN, TP]])

accuracy = (TP + TN) / (TP + TN + FP + FN)

print("\n Confusion Matrix (Manual):")
print(cm)
print(f"\n Accuracy (Manual): {accuracy * 100:.2f}%")
```

```
 Confusion Matrix (Manual):
[[ 6  5]
 [ 6 23]]

 Accuracy (Manual): 72.50%
```

- **Accuracy:**

$$\frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:**

$$\frac{TP}{TP + FP}$$

→ How many predicted positives are correct

- **Recall (Sensitivity):**

$$\frac{TP}{TP + FN}$$

→ How many actual positives are correctly detected

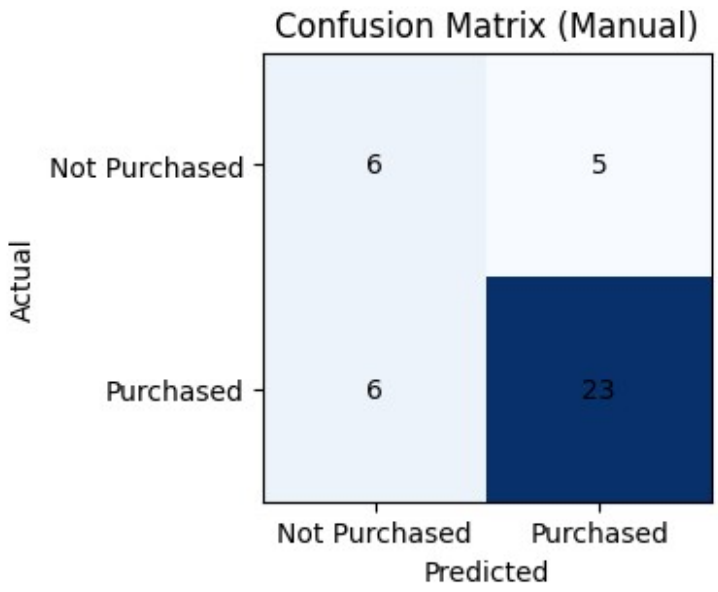
The **F1 score** is the **harmonic mean** of Precision and Recall:

$$F1 = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

It balances **false positives** and **false negatives** — very useful when your dataset is **imbalanced** (e.g., more non-purchasers than purchasers).

```
import matplotlib.pyplot as plt

# Visualize confusion matrix using matplotlib
plt.figure(figsize=(4,3))
plt.imshow(cm, cmap='Blues')
plt.title('Confusion Matrix (Manual)')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.xticks([0,1], ['Not Purchased', 'Purchased'])
plt.yticks([0,1], ['Not Purchased', 'Purchased'])
for i in range(2):
    for j in range(2):
        plt.text(j, i, cm[i,j], ha='center', va='center',
color='black')
plt.show()
```


Confusion Matrix (Manual)
Not Purchased
Purchased
Actual
6
5
6
23
Not Purchased
Purchased
Predicted